

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Which benign liver tumor is associated with oral contraceptive use in reproductive-age women, and what is its main danger?
2. What is the most sensitive clue that a liver tumor is metastatic rather than primary, based on its gross pattern?
3. Which exposures are linked to hepatic angiosarcoma?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Liver tumors — the frame

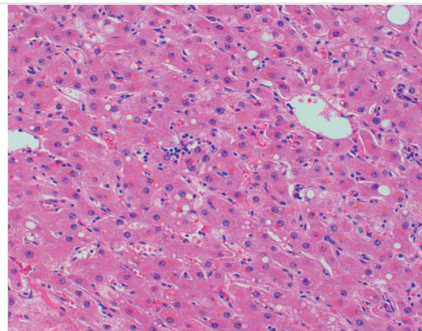
What	The liver hosts benign and malignant, primary and metastatic tumors.
Origin of primary tumors	Arise from different cellular elements: hepatocyte , biliary tree , vascular element , etc.
Presentation	Ranges from asymptomatic (found on imaging) to abdominal pain, abdominal mass, or impaired liver function (e.g. obstructive jaundice).

2 Benign liver tumors**Hepatocellular adenoma**

What	Benign tumor, usually in a non-cirrhotic liver of reproductive-age women .
Aetiology	Oral contraceptive use · obesity/metabolic syndrome · genetic mutations (e.g. β-catenin).
Clinical significance	Prone to rupture and intra-abdominal hemorrhage (risk rises with tumor size); carries a risk of malignant transformation .



Hepatocellular adenoma: well-demarcated mass, similar to background liver tissue



Hepatocellular adenoma, histology: uniform cells, minimal atypia, architecturally normal hepatic plates

Hepatic hemangioma

What	Benign vascular tumor ; the most common benign liver tumor ; solitary or multiple.
Clinical significance	Majority asymptomatic . Larger lesions (>5 cm) are liable for hemorrhage and thrombosis/consumptive coagulopathy .

3 Malignant liver tumors**Hepatocellular carcinoma (HCC)**

What	The commonest primary malignant liver tumor in adults .
Clinical	Features of underlying chronic liver disease (if present), abdominal pain, malaise, fatigue, weight loss, awareness of abdominal mass.
Tumor marker	α-fetoprotein (AFP) : elevated in only ~50% of advanced HCC — neither sensitive nor specific.

Aetiology

Cause	Detail
Chronic HBV infection	Highest in Asia (SE China, Korea, Taiwan) and sub-Saharan Africa, with vertical transmission
Hepatitis C	Increasingly important cause in the Western world
Aflatoxin	Mycotoxin from <i>Aspergillus</i> species contaminating staple crops in Africa/Asia
Alcohol	
NAFLD	
Inherited disorders	Hereditary hemochromatosis and a1AT deficiency especially; Wilson disease to a lesser degree

Pathology

Gross patterns	Microscopic
Unifocal (usually large) mass · multifocal, widely distributed nodules of variable size · diffusely infiltrative cancer	Ranges from well-differentiated (mimics normal liver) to high-grade pleomorphic tumor; pseudoacinar spaces, malformed/dilated bile canaliculi, thickened hepatocyte trabeculae



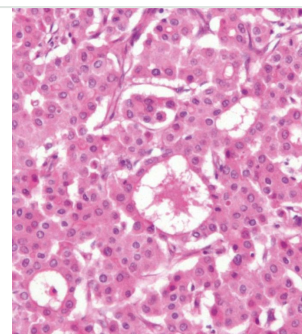
HCC: unifocal mass (circled) in a cirrhotic liver background



HCC, diffuse pattern: hundreds of small nodules, hard to distinguish from cirrhotic nodules



HCC, multifocal: widely distributed cream-tan nodules with hemorrhage



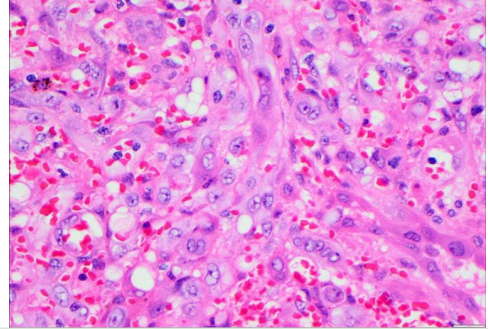
HCC histology: malignant hepatocytes with pseudoacinar spaces and thickened trabeculae

Other malignant liver tumors

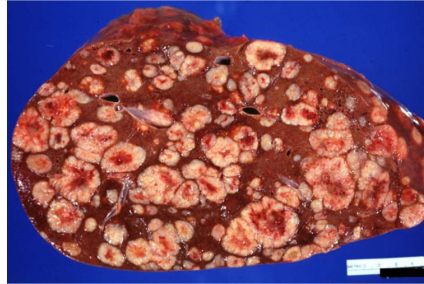
Tumor	Key features
Hepatoblastoma	Most common hepatic malignancy in infants/children ; immature precursor cells with variable epithelial and mesenchymal components
Angiosarcoma	Malignant vascular tumor; commonest primary liver sarcoma ; associated with androgen steroids, arsenic, Thorotrast, vinyl chloride
Metastatic tumors	Most common liver malignancy overall ; common primaries: colon, breast, lung ; typically multiple, well-circumscribed masses



Angiosarcoma: diffusely infiltrative, hemorrhagic, blood-filled cavitary mass



Angiosarcoma histology: malignant cells forming anastomosing vascular channels



Metastatic tumors: multiple well-circumscribed pale nodules scattered through the liver

P Must-remember pearls

- **Hepatocellular adenoma**: reproductive-age women, OCP-associated, non-cirrhotic liver — risk is **rupture/hemorrhage** and malignant transformation. **Hemangioma** = most common benign liver tumor (vascular); risk only if >5 cm.
- **HCC = commonest primary malignant liver tumor**; AFP is elevated in only ~50% of advanced cases (not sensitive/specific). Major causes: **HBV (Asia/Africa, vertical transmission), HCV (West), aflatoxin, alcohol, NAFLD, hemochromatosis, α 1AT deficiency**.
- HCC gross patterns: **unifocal, multifocal, or diffusely infiltrative**; microscopic hallmark: **pseudoacinar spaces, thickened trabeculae, malformed bile canaliculi**.
- **Hepatoblastoma = most common liver malignancy in children**. **Angiosarcoma = commonest primary liver sarcoma**, linked to Thorotrast/arsenic/vinyl chloride/androgens — presents as a hemorrhagic, blood-filled mass.
- **Metastases are the most common malignant tumor found in the liver overall** (colon, breast, lung are top primaries), classically **multiple, well-circumscribed** nodules — contrast with HCC's pseudoacinar, trabecular pattern.

Q Test yourself

1. A 28-year-old woman on long-term oral contraceptives is found to have a well-demarcated liver mass on imaging. Biopsy shows uniform hepatocytes with minimal atypia arranged in architecturally normal plates. The main clinical concern with this tumor is:

- A.** Rupture and intra-abdominal hemorrhage
- B.** Metastasis to the lung
- C.** Progression to cirrhosis
- D.** Obstructive jaundice
- E.** Portal vein thrombosis

2. Which tumor marker is used for hepatocellular carcinoma, despite being neither sensitive nor specific?

- A.** CA 19-9
- B.** CEA
- C.** PSA
- D.** Alpha-fetoprotein (AFP)
- E.** CA-125

3. A liver biopsy shows malignant hepatocytes with large pseudoacinar spaces, malformed dilated bile canaliculi, and thickened hepatocyte trabeculae. This is most consistent with:

- A.** Hepatic hemangioma
- B.** Metastatic adenocarcinoma
- C.** Hepatocellular carcinoma
- D.** Angiosarcoma
- E.** Hepatoblastoma

4. A 2-year-old presents with an abdominal mass. Biopsy shows immature precursor cells with mixed epithelial and mesenchymal components. This is most consistent with:

- A.** Hepatocellular carcinoma
- B.** Hepatic adenoma
- C.** Angiosarcoma
- D.** Metastatic neuroblastoma
- E.** Hepatoblastoma

5. A patient with a history of occupational vinyl chloride exposure develops a diffusely infiltrative, hemorrhagic, blood-filled liver mass. The most likely diagnosis is:

- A.** Hepatocellular carcinoma
- B.** Angiosarcoma
- C.** Hepatoblastoma
- D.** Hepatic hemangioma
- E.** Metastatic sarcoma

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 • A — Hepatocellular adenoma is prone to rupture causing intra-abdominal hemorrhage, with risk increasing with tumor size, and carries some risk of malignant transformation.

2 • D — Serum AFP is used as a tumor marker for HCC but is elevated in only about 50% of advanced HCC cases, making it neither sensitive nor specific.

3 • C — Pseudoacinar spaces, malformed/dilated bile canaliculi, and thickened hepatocyte trabeculae are classic histologic features of hepatocellular carcinoma.

4 • E — Hepatoblastoma is the most common hepatic malignancy in infants and children, showing immature precursor cells with variable epithelial and mesenchymal components.

5 • B — Angiosarcoma, the most common primary sarcoma of the liver, is associated with exposure to vinyl chloride, arsenic, Thorotrast, and androgenic steroids, and presents as a hemorrhagic, infiltrative mass.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from SGL5 (titled "Liver, pancreas and biliary system" — slide content covers liver only), Dr Savan Saeed Azeez, Hawler Medical University
Study alongside the lecture, not instead of it.